INCH-POUND
MS35334E
23 May 2022
SUPERSEDING
MS35334D
6 February 1974

WASHER, LOCK, FLAT, HEAVY, INTERNAL TOOTH

Reactivated after 23 May 2022 and may be used for new and existing designs and acquisitions.

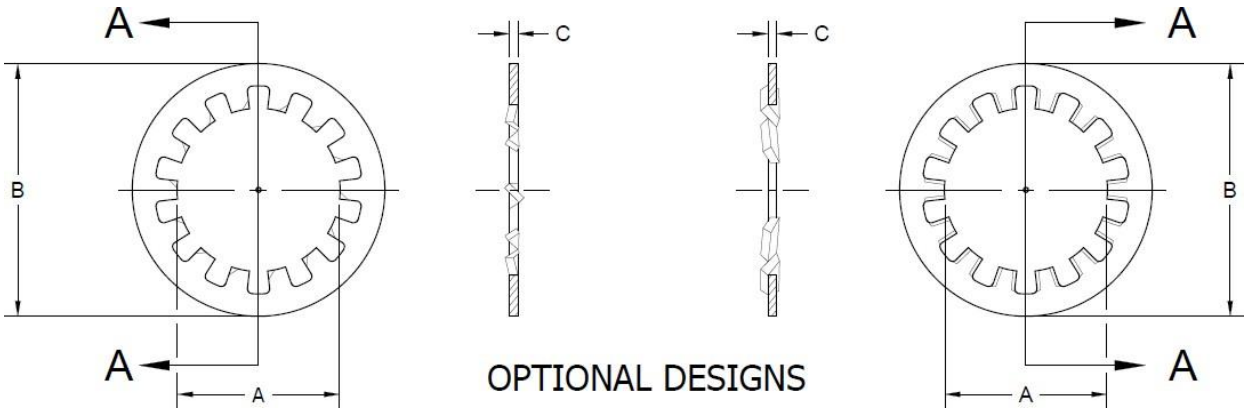


FIGURE 1. Internal tooth lock washer.

NOTE: All washers procured after the date of this specification shall be Cadmium and Hexavalent-Chromium free. Washers procured prior to the date of this specification may be use.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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TABLE I. Dash numbers (inch).

DASH NUMBERS			NOMINAL SIZE	A INSIDE DIAMETER		B OUTSIDE DIAMETER		C THICKNESS	
CADMIUM PLATE[1]	MECH. ZINC[2]	ZINC-NICKEL PLATE[3]		MAX	MIN	MAX	MIN	MAX	MIN
19	19AS	19QW	0.2500	0.267	0.256	0.536	0.500	0.045	0.035
20	20AS	20QW	0.3125	0.332	0.320	0.607	0.590	0.050	0.040
21	21AS	21QW	0.3750	0.398	0.384	0.748	0.700	0.050	0.042
22*	*	*	0.4375	0.464	0.448	0.858	0.800	0.067	0.050
23	23AS	23QW	0.5000	0.530	0.512	0.924	0.880	0.067	0.055
24*	*	*	0.5625	0.596	0.576	1.034	0.990	0.067	0.055
25	25AS	25QW	0.6250	0.663	0.640	1.135	1.100	0.067	0.059
26	26AS	26QW	0.7500	0.795	0.768	1.265	1.240	0.084	0.070
27	27AS	27QW	0.8750	0.927	0.894	1.447	1.400	0.084	0.075

NOTE: [1] Do not use for new design. Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List.
[2] Preferred for elimination of hydrogen embrittlement risk during the finish application process.
[3] Only use for applications where mechanical zinc finish does not provide adequate corrosion protection.
* Inactive for new design after 6 February 1974. There is no direct compatible translation from Inch to Metric, thus, Table II from previous issue has been deleted.

REQUIREMENTS:

1 Material. Unless otherwise specified, washers shall be made from steel, SAE J403 1050–1065 (UNS G10500–G10650) per ASME B18.21.1.

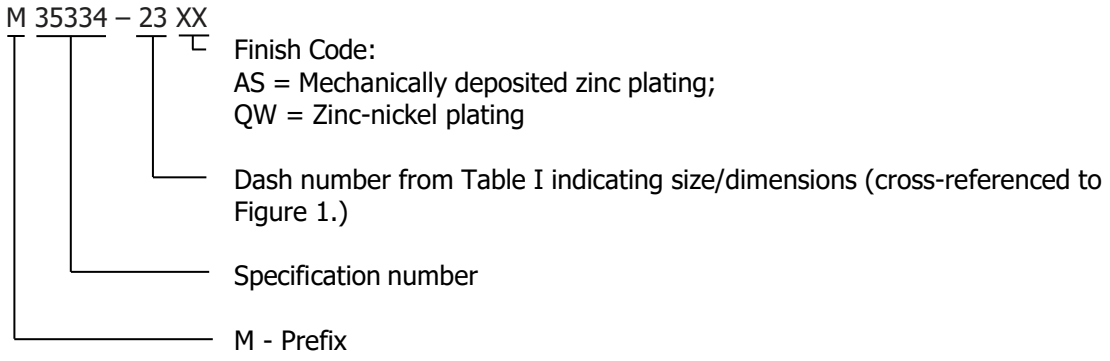
2 Hardness. Washers shall be quenched and tempered to HRC 40 – 50, HRC, 392HV to 513HV per ASME B18.21.1.

3 Protective finish. Add one of the following codes as specified in the contract, after the dash number to designate protective finish:

AS - Zinc Mechanically Deposited per ASTM B695, Type II, Class 5 with conversion treatment free of hexavalent chromium and meets the Type II corrosion resistance.
QW - Zinc-Nickel plate, Type K20 per MIL-PRF-32647.

4 Conformance. Certificates of compliance to paragraphs 1, 2, and 3 shall be provided.

5 Part identifying number. The PIN consists of the following:



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Example: MS35334-23AS is a washer, lock-flat, heavy, internal tooth, mechanically deposited zinc coated having a nominal size of 0.5000 with an ID of between 0.512 - 0.530, OD of between 0.880 - 0.924 and a thickness of between 0.055 - 0.067.

MS35334-19QW is a washer, lock-flat, heavy, internal tooth, zinc-nickel plated having a nominal size of 0.2500 with an ID of between 0.256 – 0.267, OD of between 0.500 – 0.536 and a thickness of between 0.035 – 0.045.

6 Washer teeth. The number of teeth shall be at the option of the manufacturer. The teeth shall be uniform and symmetrical with respect to size, shape, and angle of twist or of bent edges.

7 Documents. Referenced documents shall be of the issue in effect on the date of the invitation for bids.

8 Design. For design feature purposes, this document takes precedence over all other documents.

9 Documents. Referenced documents shall be of the issue in effect on the date of the invitation for bids. Copies of FF-W-100 and MIL-PRF-32647 are available online at https://quicksearch.dla.mil. Copies of ASTM documents are available online at https://www.astm.org/Standard/standards-and-publications. Copies of ASME documents are available online at https://www.asme.org.

CONCLUDING MATERIAL

Custodian:
Army – AT
DLA – GS

Preparing Activity:
Army – AT

(Project 5310-2022-009)

Review Activity:
Air Force – 71

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